Uniform thermal information for bounded-degree quantum interactions

A localized-adjoint anti-cancellation argument

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Abstract

For a canonical family of distinct nonidentity Pauli interactions of bounded support and bounded incidence per site, we derive a lower bound on the centered KMS quadratic form of every linear combination, uniform in system size and valid at every finite temperature. With maximum incident interaction strength g, the lower bound has the form $\exp[-C_{k,\Delta}(1+\beta g)^2]\|u\|_2^2$. It therefore also bounds the Bogoliubov–Kubo–Mori covariance and the Hessian of the log-partition function. The mechanism combines local spectral multiplicity with Gaussian-smoothed commutator gradients. Their exact thermal adjoints remain quasi-local; transferring the adjoint before estimating creates a summable Gram kernel without any assumption of thermal correlation decay. We also prove that bounded interaction strength without bounded combinatorial incidence is insufficient, even for classical two-body Hamiltonians with all candidate interactions nonzero. The inverse-response and statistical consequences are stated separately from computational and state-preparation costs.

Status. This is a newly derived proof manuscript, not an externally reviewed or formally verified result. The mathematics is supplied below, including the local input. The targeted literature comparison is not a certification of original priority. Numerical checks in the companion packet test finite instances and algebraic identities only.

1 Statement and exact scope

Let $\mathcal{P} = \{P_1, \dots, P_m\}$ be distinct, nonidentity elements of $\{I, X, Y, Z\}^{\otimes n}$, with their canonical signs. Put $S_a = \operatorname{supp} P_a$. Assume

$$1 \le |S_a| \le k, \qquad \max_i \#\{a : i \in S_a\} \le \Delta, \qquad H(h) = \sum_a h_a P_a, \qquad \max_i \sum_{a: i \in S_a} |h_a| \le g. \tag{1}$$

The incidence bound applies to the entire candidate family, including terms whose coefficient happens to vanish. We take $g > 0$; $g = 0$ is trivial. At a known inverse temperature $\beta > 0$, let $\rho = e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$. For $A = A^\dagger$, write $A_c = A - \operatorname{Tr}(\rho A) I$, and define

$$\langle X, Y \rangle_\rho = \operatorname{Tr}(\rho^{1/2} X^\dagger \rho^{1/2} Y), \tag{2}$$

$$\operatorname{Cov}_\rho^{\mathrm{BKM}}(A, A) = \int_0^1 \operatorname{Tr}(\rho^t A_c \rho^{1-t} A_c)\, dt. \tag{3}$$

Both traces are ordinary, unnormalized matrix traces; $\operatorname{Tr}\rho = 1$. We use $\tau(X) = 2^{-n}\operatorname{Tr} X$ for normalized trace.

Theorem 1.1 (Uniform thermal conditioning)**.** *For each fixed k, Δ there is a finite constant $C_{k,\Delta}$ such that every system satisfying* (1)*, every $u \in \mathbb{R}^m$, and $A = \sum_a u_a P_a$ satisfy*

$$\mathrm{Cov}_\rho^{\mathrm{BKM}}(A, A) \;\geq\; \|A_c\|^2_{\rho,\mathrm{KMS}} \;\geq\; e^{-C_{k,\Delta}(1+\beta g)^2} \sum_a u_a^2. \tag{4}$$

The constant is independent of n, m, the graph, the coefficient signs, spectral gaps, and mixing times. A construction of the constant is given in Section 7. At $\beta = 0$ the two quadratic forms equal $\sum_a u_a^2$ exactly.

No commutativity, lack of frustration, high-temperature condition, uniqueness of a thermodynamic phase, or decay of correlations is assumed. The state is the unrestricted Gibbs state on the full qubit tensor product. A Gibbs state projected onto a fixed conserved-charge sector is a different object and is not covered. Fixed-locality models with growing candidate incidence are also not covered.

Rescale $\widehat{H} = H/g$ and $b = \beta g$. Henceforth the maximum incident strength of $\widehat{H}$ is at most one and $\rho = e^{-b\widehat{H}}/Z$. Connect two sites whenever they share a candidate support. This graph has degree at most $q = \Delta(k-1)$, and every S_a has diameter at most one. Its balls obey

$$|B(i, r)| \leq v_0 e^{\nu r}, \qquad v_0 = q + 2, \quad \nu = \log(q + 2). \tag{5}$$

Set dist $= \infty$ across disconnected components. For $\lambda > \nu$, let

$$V_\lambda = \frac{v_0}{1 - e^{\nu-\lambda}}; \qquad \sup_i \sum_x e^{-\lambda \,\mathrm{dist}(i,x)} \leq V_\lambda. \tag{6}$$

Constants denoted C may depend on k, Δ and on explicitly fixed spatial exponents, never on n, m, b.

2 The local spectral-multiplicity input

We include the proof rather than assuming an unreviewed earlier claim.

Lemma 2.1 (Local thermal faithfulness)**.** *Let $H_* = I_S \otimes K + V$, where $\dim \mathcal{H}_S = d$, $K = K^\dagger$, $V = V^\dagger$ and $\|V\|_{\mathrm{op}} \leq v$. Let $\rho_* = e^{-\beta H_*}/Z_*$. For every $X = x \otimes I$, including non-Hermitian x,*

$$\|X\|^2_{\rho_*,\mathrm{KMS}} := \mathrm{Tr}(\rho_*^{1/2} X^\dagger \rho_*^{1/2} X) \geq \frac{4}{9d} e^{-4\beta v} \|x\|^2_{\mathrm{op}} \geq \frac{4}{9d} e^{-4\beta v} \frac{\mathrm{Tr}_S\, x^\dagger x}{d}. \tag{7}$$

There is no bound on the dimension or energy scale of the bath.

Proof. First, if A, B are Hermitian, $\|A - B\|_{\mathrm{op}} \leq v$, $P_0 = \mathbf{1}_{(-\infty,E]}(B)$ and $Q = \mathbf{1}_{(E+t,\infty)}(A)$, then

$$\|QP_0\|_{\mathrm{op}} \leq v/t \quad (t > 0). \tag{8}$$

Indeed, $T = QP_0$ solves $A_Q T - T B_{P_0} = Q(A - B)P_0$. On the two compressed subspaces,

$$T = \int_0^\infty e^{-s(A_Q - EI)} Q(A - B) P_0 e^{s(B_{P_0} - EI)} ds,$$

whose norm is at most $\int_0^\infty v e^{-st} ds$.

If $P = I - Q$ and $[X, P_0] = 0$, expanding $P_0 PXPP_0 - P_0 X P_0$ gives

$$\|P_0 PXPP_0 - P_0 X P_0\|_{\mathrm{op}} \leq 3 \|QP_0\|^2_{\mathrm{op}} \|X\|_{\mathrm{op}}. \tag{9}$$

The three terms are $-P_0QXP_0$, $-P_0XQP_0$, and P_0QXQP_0; commutation with P_0 makes each bound quadratic in $\|QP_0\|_{\mathrm{op}}$.

Apply these observations with $B = I \otimes K$, $A = H_*$, and $t = 3v$. If $r(E) = \operatorname{rank} \mathbf{1}_{(-\infty,E]}(K)$, the compression P_0XP_0 contains $r(E)$ copies of x. The singular-value perturbation bound therefore leaves at least $r(E)$ singular values of P_0PXPP_0 of size at least $2\|x\|_{\mathrm{op}}/3$. Thus, with $P_H(t) = \mathbf{1}_{(-\infty,t]}(H_*)$,

$$F(E+3v) := \|P_H(E+3v)XP_H(E+3v)\|_F^2 \geq \tfrac{4}{9} r(E)\|x\|_{\mathrm{op}}^2.$$

In an energy eigenbasis,

$$\int_{\mathbb{R}} \beta e^{-\beta t} F(t)\,dt = \sum_{ij} e^{-\beta \max(E_i,E_j)} |X_{ij}|^2 \leq Z_* \|X\|_{\rho_*,\mathrm{KMS}}^2.$$

The integral of $\beta e^{-\beta E} r(E)$ is $Z_B = \operatorname{Tr} e^{-\beta K}$. The min–max eigenvalue inequality gives the scalar comparison $Z_* \leq d e^{\beta v} Z_B$. These facts prove (7). If $v = 0$ the Gibbs state factorizes and the conclusion follows directly. No operator-order comparison of noncommuting Gibbs exponentials is used. □

Let $Q_{i,\alpha}$ denote the one-site Pauli $\alpha \in \{X, Y, Z\}$. The support of $[Q_{i,\alpha}, A]$ has at most

$$s = 1 + \Delta(k-1)$$

sites. The sum of Hamiltonian terms touching this support has norm at most s. Lemma 2.1 gives

$$\|[Q_{i,\alpha}, A]\|_{\rho,\mathrm{KMS}}^2 \geq \kappa_b \tau([Q_{i,\alpha}, A]^\dagger [Q_{i,\alpha}, A]), \qquad \kappa_b = \frac{4}{9\,2^s} e^{-4bs}. \tag{10}$$

For fixed (i, α), left multiplication by $Q_{i,\alpha}$ preserves Pauli orthogonality. Exactly two of the three one-site Paulis anticommute with each nonidentity factor. Hence

$$S(A) := \sum_{i,\alpha} \|[Q_{i,\alpha}, A]\|_{\rho,\mathrm{KMS}}^2 \geq 8\kappa_b \sum_a |S_a| u_a^2 \geq 8\kappa_b \|u\|_2^2. \tag{11}$$

This does not yet lower-bound $\|A_c\|_{\rho,\mathrm{KMS}}^2$: controlling the sum without losing a factor m is the global issue.

3 Gaussian gradients and their exact thermal adjoints

For $j = (i, \alpha)$ and real ω, define

$$B_{j,\omega} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-t^2/2} e^{-i\omega t} e^{it\widehat{H}} Q_j e^{-it\widehat{H}}\,dt. \tag{12}$$

Then $\|B_{j,\omega}\|_{\mathrm{op}} \leq 1$, $B_{j,\omega}^\dagger = B_{j,-\omega}$, and

$$e^{z\widehat{H}} B_{j,\omega} e^{-z\widehat{H}} = e^{z\omega + z^2/2} B_{j,\omega+z} \qquad (z \in \mathbb{R}). \tag{13}$$

To verify the last equality, the (r, s) matrix element of $B_{j,\omega}$ is $(Q_j)_{rs} \exp[-(\omega - (E_r - E_s))^2/2]$; complete the square. Also

$$Q_j = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} B_{j,\omega}\,d\omega. \tag{14}$$

Absolute operator-norm convergence follows from the frequency estimate in Section 4.

Put $\partial_{j,\omega}(X) = [B_{j,\omega}, X]$. For the KMS inner product its exact adjoint is

$$\begin{aligned}\partial^{\sharp}_{j,\omega}(Y) &= \rho^{-1/2} B^{\dagger}_{j,\omega} \rho^{1/2} Y - Y \rho^{1/2} B^{\dagger}_{j,\omega} \rho^{-1/2} \\ &= e^{-b\omega/2 + b^2/8} B_{j,-\omega+b/2} Y - e^{b\omega/2 + b^2/8} Y B_{j,-\omega-b/2}.\end{aligned} \tag{15}$$

The first line follows by inserting $\Omega_\rho(Y) = \rho^{1/2} Y \rho^{1/2}$ into $\Omega_\rho^{-1} \partial^{\dagger}_{\mathrm{HS}} \Omega_\rho$; the second follows from (13). In particular, no uncontrolled full-system imaginary-time operator norm occurs.

Choose

$$w_b(\omega) = e^{-(1+b)|\omega|}, \qquad G_j = \int_{\mathbb{R}} w_b(\omega) \partial^{\sharp}_{j,\omega} \partial_{j,\omega} \, d\omega, \qquad G = \sum_j G_j. \tag{16}$$

Each G_j is KMS-self-adjoint, KMS-positive, and annihilates I. Moreover

$$D(A) := \langle A, GA \rangle_\rho = \sum_j \int_{\mathbb{R}} w_b(\omega) \| [B_{j,\omega}, A] \|^2_{\rho,\mathrm{KMS}} d\omega, \tag{17}$$

$$\|G_j\|_{\infty \to \infty} \le 8 e^{b^2/8}. \tag{18}$$

These follow directly from (15). The frequency integrand for the norm bound is at most

$$4 e^{b^2/8} e^{-(1+b/2)|\omega|}.$$

Why no sampling generator is required. G is an auxiliary positive operator on the KMS Hilbert space. We do not require that $-G$ generate a completely positive semigroup, nor do we prove a spectral gap on the full operator algebra. Dropping that unnecessary requirement allows a Gaussian with a fixed time width and arbitrarily strong spatial decay, at a finite temperature-dependent prefactor. Gaussian operator Fourier transforms and exact thermal adjoints have important prior precedents in Gibbs sampling and local learning [8, 5, 6]; their use here must not be described as entirely new in isolation.

4 Spatial and frequency tails

Let $\widehat{H}_i^{(r)}$ retain only Hamiltonian terms whose supports are wholly inside $B(i, r)$. Let $B^{(r)}_{j,\omega}$ be (12) for this truncated Hamiltonian, embedded into the full tensor product.

Lemma 4.1 (Uniform filtered locality)**.** *For each fixed $\lambda > 0$ there is a constant $C_\lambda = C_\lambda(k, \Delta)$ such that*

$$\sup_\omega \| B_{j,\omega} - B^{(r)}_{j,\omega} \|_{\mathrm{op}} \le C_\lambda e^{-\lambda r}. \tag{19}$$

The exponent λ can be arbitrarily large, with a correspondingly large but system-size-independent prefactor.

Proof. Here is the finite-range Lieb–Robinson estimate needed, including the reason it holds on exponentially growing graphs. The usual commutator iteration (obtained by differentiating a commutator and removing the terms supported on its initial region) gives a sum over chains of consecutively intersecting interactions. For an initial one-site operator, the total weight of chains of length $\ell \ge 1$ is at most $k^{\ell-1}$: the first incident interaction has total norm at most one, and every

subsequent interaction intersects at most k sites of its predecessor. A chain reaching distance R has length at least R. Consequently, for any $\lambda' > 0$,

$$\|[e^{it\widehat{H}'}Q_j e^{-it\widehat{H}'}, O_Y]\|_{\mathrm{op}} \le 2\|O_Y\|_{\mathrm{op}} e^{-\lambda' \operatorname{dist}(i,Y)} e^{2ke^{\lambda'}|t|},$$

with harmless enlargement at distance zero. The same estimate holds for every truncated Hamiltonian $\widehat{H}'$; this is the standard finite-range commutator-path proof of the Lieb–Robinson bound [9].

Duhamel's formula compares full and ball-restricted evolution of Q_j. Only interaction terms crossing the ball boundary contribute. Each lies at distance at least r from i, and their total norm is at most $v_0 e^{\nu r}$. With $\lambda' = \lambda + \nu$ the resulting difference is bounded by

$$\frac{2v_0}{v_{\lambda'}} e^{-\lambda r} e^{v_{\lambda'}|t|}, \qquad v_{\lambda'} = 2ke^{\lambda'}.$$

Integrate this bound against the Gaussian; $\int (2\pi)^{-1/2} e^{-t^2/2} e^{v|t|} dt \le 2e^{v^2/2}$. This proves (19). For example, increasing constants if necessary,

$$C_\lambda = 4v_0 \exp\Big(2k^2 e^{2(\lambda+\nu)}\Big) \tag{20}$$

is sufficient. Its size is not claimed practical. The Fourier phase has modulus one, so the estimate is uniform in ω. □

Define $G_j^{(r)}$ by replacing every B in the explicit second line of (15) and in ∂ by $B^{(r)}$. This is a linear map acting only on $B(i,r)$, it annihilates the identity, and it annihilates any observable supported disjointly from that ball. It need not be self-adjoint for the full-state inner product. From (19) and the same integrable frequency weight,

$$\|G_j\|_{\infty\to\infty},\ \|G_j^{(r)}\|_{\infty\to\infty} \le M_b, \qquad \|G_j - G_j^{(r)}\|_{\infty\to\infty} \le M_b e^{-\mu r}, \quad M_b = (8 + 16C_\mu)e^{b^2/8}. \tag{21}$$

We will fix $\mu = 20(1+\nu)$.

Lemma 4.2 (Joint spatial–frequency tails)**.** *Let $a_0 = (4k)^{-1}$ and $a = a_0/2$. With $B^{(-1)} = 0$, define $E_{j,r,\omega} = B_{j,\omega}^{(r)} - B_{j,\omega}^{(r-1)}$. There is a constant C_E depending only on k, Δ, μ such that*

$$\|E_{j,r,\omega}\|_{\mathrm{op}} \le C_E e^{-\mu r} e^{-a|\omega|}. \tag{22}$$

Every shell is supported on $B(i,r)$.

Proof. Expanding nested commutators starting at one site gives

$$\|\operatorname{ad}_{\widehat{H}'}^{\ell}(Q_j)\|_{\mathrm{op}} \le (2k)^\ell \ell!$$

for both full and truncated Hamiltonians: after r commutators a term has support size at most $1 + r(k-1)$, and the total norm of interactions incident on that support is at most this number. Therefore

$$\|e^{z\widehat{H}'} Q_j e^{-z\widehat{H}'}\|_{\mathrm{op}} \le 2 \qquad (|z| \le a_0).$$

Complete the Gaussian square, now with $z = a_0 \operatorname{sign}\omega$, to obtain

$$\|B_{j,\omega}^{(r)}\|_{\mathrm{op}},\ \|B_{j,\omega}\|_{\mathrm{op}} \le 2e^{a_0^2/2} e^{-a_0|\omega|}. \tag{23}$$

The spatial bound at exponent 2μ bounds the shell by $C_{2\mu}(1 + e^{2\mu})e^{-2\mu r}$. The frequency bound is $4e^{a_0^2/2}e^{-a_0|\omega|}$. Taking the geometric mean of these two valid bounds proves (22); enlarge the constant to include $r = 0$. □

For $\Omega \geq 0$, set

$$R_{j,\Omega} = Q_j - \frac{1}{\sqrt{2\pi}} \int_{-\Omega}^{\Omega} B_{j,\omega} d\omega.$$

Integrating the shells outside $[-\Omega, \Omega]$ yields an absolutely convergent decomposition $R_{j,\Omega} = \sum_{r \geq 0} R_{j,r,\Omega}$ with

$$\operatorname{supp} R_{j,r,\Omega} \subset B(i,r), \qquad \|R_{j,r,\Omega}\|_{\mathrm{op}} \leq C_R e^{-a\Omega} e^{-\mu r}, \quad C_R = \frac{2C_E}{a\sqrt{2\pi}}. \tag{24}$$

Thus

$$\|[R_{j,\Omega}, A]\|_{\mathrm{op}} \leq \frac{2C_R}{1-e^{-\mu}} e^{-a\Omega} \sum_a e^{-\mu \operatorname{dist}(i,S_a)} |u_a|.$$

The nonnegative kernel $K_{ia} = e^{-\mu \operatorname{dist}(i,S_a)}$ has row sums at most ΔV_μ and column sums at most kV_μ. Accounting for the three Paulis per site and using the Schur test, we obtain the important *coefficient* ℓ_2 tail estimate

$$\sum_j \|[R_{j,\Omega}, A]\|_{\rho,\mathrm{KMS}}^2 \leq T^2 e^{-2a\Omega} \|u\|_2^2, \quad T = \max\left\{1, \frac{2C_R\sqrt{3k\Delta}V_\mu}{1-e^{-\mu}}\right\}. \tag{25}$$

We used $\|X\|_{\rho,\mathrm{KMS}} \leq \|X\|_{\mathrm{op}}$, which follows from Schatten Hölder. Replacing (25) by a bound involving $n\|u\|_\infty^2$ would not prove the theorem.

5 Coercivity of the auxiliary quadratic form

Use the decomposition into $|\omega| \leq \Omega$ and its residual in the Hilbert-space direct sum over j. Minkowski's and Cauchy–Schwarz inequalities give

$$\sqrt{S(A)} \leq L_{b,\Omega}\sqrt{D(A)} + Te^{-a\Omega}\|u\|_2, \qquad L_{b,\Omega}^2 = \frac{1}{2\pi}\int_{-\Omega}^{\Omega} w_b(\omega)^{-1} d\omega = \frac{e^{(1+b)\Omega}-1}{\pi(1+b)}. \tag{26}$$

Choose

$$\Omega_b = \max\left\{1, \frac{1}{a}\log\frac{2T}{\sqrt{8\kappa_b}}\right\}. \tag{27}$$

By (11), the residual is at most half the lower bound on $\sqrt{S(A)}$. Consequently

$$D(A) \geq \eta_b \|u\|_2^2, \qquad \eta_b = \frac{2\kappa_b}{L_{b,\Omega_b}^2}. \tag{28}$$

Here $\Omega_b = O_{k,\Delta}(1+b)$ and $\eta_b \geq \exp[-C_{k,\Delta}(1+b)^2]$ after increasing the constant.

6 The global anti-cancellation lemma

The next lemma is the second essential step. It concerns correlations between transformed observables, not correlations between the original local observables.

Lemma 6.1 (Localized-adjoint image bound)**.** *Let a graph satisfy* (5)*, and let an inner product* $\langle\cdot,\cdot\rangle_*$ *on operators obey* $\|X\|_* \leq \|X\|_{\mathrm{op}}$*. Let there be at most* r_0 *maps* F_j *centered at each vertex,*

each exactly self-adjoint in that inner product. Assume local maps $F_j^{(r)}$ act only on $B(i,r)$, annihilate every operator supported disjointly from the ball, and satisfy

$$\|F_j\|_{\infty\to\infty},\ \|F_j^{(r)}\|_{\infty\to\infty} \le M, \qquad \|F_j - F_j^{(r)}\|_{\infty\to\infty} \le M e^{-\mu r}.$$

For local operators P_a of norm at most one and support size at most k, with incidence at most Δ, if $\mu > 8\nu$ then

$$\left\| \sum_j F_j \Big(\sum_a u_a P_a \Big) \right\|_*^2 \le C_* M^2 \sum_a |u_a|^2. \tag{29}$$

The constant depends only on $k, \Delta, r_0, \mu, v_0, \nu$.

Proof. Write $F = \sum_j F_j$. If $d = \operatorname{dist}(i, S_a) > 0$, then $F_j^{(d-1)} P_a = 0$, so for all $d \ge 0$,

$$\|F_j P_a\|_{\mathrm{op}} \le M e^{\mu} e^{-\mu \operatorname{dist}(i,S_a)}. \tag{30}$$

It follows that $\sum_j \|F_j P_a\|_{\mathrm{op}} \le A_0 M$ and

$$\sum_{j:\operatorname{dist}(i,S_a)>L} \|F_j P_a\|_{\mathrm{op}} \le A_1 M e^{-(\mu-\nu)L},$$

where one can take

$$A_0 = r_0 k e^{\mu} V_\mu, \qquad A_1 = \frac{r_0 k v_0 e^{\mu}}{1 - e^{-(\mu-\nu)}}.$$

Consider the Gram matrix

$$K_{ac} = \langle F P_a, F P_c \rangle_* = \sum_{i,\alpha,j,\gamma} \langle F_{i,\alpha} P_a, F_{j,\gamma} P_c \rangle_*.$$

Put $R = \operatorname{dist}(S_a, S_c)$. For $R \ge 1$, choose $L = \lfloor R/6 \rfloor$. Terms in which i is farther than L from S_a, or j farther than L from S_c, have total absolute value at most

$$2 A_0 A_1 M^2 e^{-(\mu-\nu)L}$$

by (30) and $\|\cdot\|_* \le \|\cdot\|_{\mathrm{op}}$.

For every remaining pair, use *exact self-adjointness before estimating*:

$$\langle F_{i,\alpha} P_a, F_{j,\gamma} P_c \rangle_* = \langle P_a, F_{i,\alpha} F_{j,\gamma} P_c \rangle_*. \tag{31}$$

The support of $F_{j,\gamma}^{(L)} P_c$ is contained in $N_{2L}(S_c)$, while $F_{i,\alpha}^{(L)}$ acts within $N_{2L}(S_a)$. Since $4L < R$, these are disjoint and

$$F_{i,\alpha}^{(L)} F_{j,\gamma}^{(L)} P_c = 0.$$

The error in replacing the product by its local approximants has norm at most $2M^2 e^{-\mu L}$. There are at most $(r_0 k v_0)^2 e^{2\nu L}$ such close pairs. Therefore

$$|K_{ac}| \le \Big(2 A_0 A_1 + 2 (r_0 k v_0)^2 \Big) M^2 e^{-(\mu-2\nu)L}.$$

For $R = 0$ the direct bound is $A_0^2 M^2$. Thus, with

$$\sigma = (\mu - 2\nu)/6 > \nu,$$

there is a constant D_* such that $|K_{ac}| \le D_* M^2 e^{-\sigma \operatorname{dist}(S_a,S_c)}$. For example,

$$D_* = \max\{A_0^2,\ e^{\mu-2\nu}[2A_0A_1 + 2(r_0kv_0)^2]\}$$

suffices. For every a,

$$\sum_c e^{-\sigma \operatorname{dist}(S_a,S_c)} \le \sum_{x\in S_a}\sum_c\sum_{y\in S_c} e^{-\sigma \operatorname{dist}(x,y)} \le k\Delta V_\sigma.$$

The Schur test gives (29) with $C_* = D_* k\Delta V_\sigma$. Disconnected components cause no problem: the maps have exact component support, and the transferred product vanishes across components. □

What this does and does not assert. The proof does not bound $\mathrm{Cov}_\rho(P_a, P_c)$ by a decaying function of distance. Instead, it bounds the Gram kernel of GP_a and GP_c by transferring a localized thermal adjoint. Ordinary long-range correlations therefore do not invalidate this argument. Nor does the lemma assert a uniform norm bound for G on all operators; it controls its image on a bounded-incidence local family.

7 Proof of the main theorem and constant recipe

Apply Lemma 6.1 to $F_j = G_j$, $r_0 = 3$, and the KMS inner product, using (21). Set

$$B_0 = C_*(8 + 16C_\mu)^2.$$

Then

$$\|GA\|^2_{\rho,\mathrm{KMS}} \le B_0 e^{b^2/4}\|u\|_2^2. \tag{32}$$

Since $GI = 0$ and G is KMS-self-adjoint, $D(A) = \langle A_c, GA\rangle_\rho$. Combining (28), Cauchy–Schwarz, and (32) yields

$$\|A_c\|^2_{\rho,\mathrm{KMS}} \ge c_b\|u\|_2^2, \qquad c_b = \frac{4\kappa_b^2}{B_0 e^{b^2/4} L^4_{b,\Omega_b}} > 0. \tag{33}$$

All quantities in this expression have now been constructed using only b, k, Δ: choose the graph constants in (5), $\mu = 20(1+\nu)$, the explicit C_λ in (20), a shell constant from Lemma 4.2, then $T, \Omega_b, L_{b,\Omega_b}, C_*$ in order. Because $-\log\kappa_b = O(1+b)$, $\Omega_b = O(1+b)$ and $\log L^2_{b,\Omega_b} = O((1+b)^2)$, a finite $C_{k,\Delta}$ gives

$$c_b \ge e^{-C_{k,\Delta}(1+b)^2}.$$

These constants are not optimized and may be extremely large before their appearance in the exponent.

Finally, in an energy eigenbasis the BKM and KMS coefficients of $|(A_c)_{ij}|^2$ are respectively the logarithmic and geometric means of p_i, p_j. The logarithmic mean is at least the geometric mean. Thus BKM dominates centered KMS and Theorem 1.1 follows. □

8 Assumption boundaries and counterexamples

8.1 Bounded weighted strength does not replace bounded incidence

Proposition 8.1 (Classical star obstruction)**.** *Fix $\beta, h, \lambda > 0$. There is a family of classical two-local Pauli Hamiltonians with uniformly bounded incident interaction strength and a sequence of coefficient vectors of norm bounded away from zero whose BKM variance tends to zero.*

Proof. On $N+1$ spins put

$$H_N = -h\sum_{i=0}^{N} Z_i - \frac{\lambda}{N} Z_0 \sum_{j=1}^{N} Z_j, \qquad t = \tanh(\beta h), \qquad A_N = \frac{1}{N}\sum_{j=1}^{N} Z_0 Z_j - tZ_0.$$

All Hamiltonian coefficients in this candidate family are nonzero. The incident strength is at most $h+\lambda$, and locality is two. The coefficient norm of A_N is $t^2 + 1/N$. Conditional on $Z_0 = s \in \{-1, 1\}$, the leaves are independent and have mean

$$t_s = \tanh(\beta(h + \lambda s/N)), \qquad |t_s - t| \le \beta\lambda/N.$$

Writing $\overline{Z} = N^{-1}\sum_j Z_j$, $A_N = Z_0(\overline{Z} - t)$ and hence

$$\mathrm{Cov}^{\mathrm{BKM}}_{\rho_N}(A_N, A_N) = \mathrm{Var}_{\rho_N}(A_N) \le \mathbb{E}(\overline{Z} - t)^2 \le \frac{1}{N} + \frac{(\beta\lambda)^2}{N^2}.$$

The ratio to $t^2 + 1/N$ tends to zero. The center participates in N pair terms, so the family violates bounded incidence, exactly as required. □

For $\lambda = 0$ the same construction uses a product Gibbs state and zero candidate pair coefficients; its variance is exactly $(1 - t^2)/N$. The nonzero-coupling version avoids relying on unused candidate directions.

8.2 Low temperature and fixed-charge sectors

A single qubit with $H = hZ$ and $A = Z$ has BKM variance $\mathrm{sech}^2(\beta h) \sim 4e^{-2\beta h}$. Thus temperature-independent, or merely polynomial-in-β, lower bounds cannot hold in this class. The quadratic exponent in Theorem 1.1 is not claimed sharp.

If the state is projected onto a sector where $\sum_i Z_i$ is fixed, that observable has exactly zero variance. Such a state is not a full-rank unrestricted Gibbs state on the original tensor product, and the local Pauli family is no longer independent modulo constants on the restricted space. The theorem cannot be transferred to hard constraints or fixed-charge ensembles without quotienting these null directions.

Frustration and noncommuting interactions are allowed throughout the proof. It imposes no upper bound on the largest covariance eigenvalue, so the lower bound neither excludes phase-dependent enhancements of response nor supplies uniform susceptibility upper bounds. No zero-temperature limit uniform in β is claimed.

9 Hamiltonian identification and resource accounting

Let $\mathcal{C}$ be a convex coefficient domain obeying the common bound g in (1), and let $c = e^{-C_{k,\Delta}(1+\beta g)^2}$. Write

$$F(h) = \log \mathrm{Tr}\, e^{-\beta H(h)}, \qquad m_a(h) = \mathrm{Tr}(\rho_h P_a).$$

Direct differentiation gives $\nabla F(h) = -\beta m(h)$ and

$$\nabla^2 F(h) = \beta^2 \mathcal{I}^{\mathrm{BKM}}(h) \succeq \beta^2 c I_m. \tag{34}$$

Integrating along a parameter segment yields

Corollary 9.1 (Uniform inverse response and relative entropy)**.** *For $h, h' \in \mathcal{C}$,*

$$\|h - h'\|_2 \le \frac{1}{\beta c}\|m(h) - m(h')\|_2, \tag{35}$$

$$D(\rho_h \| \rho_{h'}) \ge \frac{\beta^2 c}{2}\|h - h'\|_2^2. \tag{36}$$

Proof. Strong convexity implies $-\beta(h-h')\cdot(m(h)-m(h')) \ge \beta^2 c\|h-h'\|_2^2$; apply Cauchy–Schwarz. Relative entropy is the Bregman divergence $F(h') - F(h) - \nabla F(h)\cdot(h'-h)$, so the second assertion follows by integrating the Hessian twice. □

This identifies all coefficients jointly from the original local moment vector, not from an enlarged set of test observables. It presupposes the candidate family and inverse temperature, and it does not identify an unknown external bath from subsystem moments alone.

9.1 Copies and measurement circuits

Color the overlap graph of candidate terms. Its maximum degree is at most $k(\Delta - 1)$, so $\chi \le k(\Delta - 1) + 1$ colors suffice. Terms in one color have disjoint supports and can be measured simultaneously with single-qubit Pauli-basis measurements. With T independent copies for each color, Hoeffding's inequality and a union bound give componentwise moment error at most δ with failure probability p using

$$N = O\left(\chi \delta^{-2} \log \frac{2m}{p}\right)$$

copies. Correlations between different terms on the same copy do not invalidate this bound.

Let $\widehat{h}$ minimize $F(x) + \beta \widehat{m}\cdot x$ over $\mathcal{C}$. The variational inequality and strong monotonicity give

$$\|\widehat{h} - h\|_2 \le \frac{\|\widehat{m} - m(h)\|_2}{\beta c}.$$

Thus total coefficient error $\|\widehat{h} - h\|_2 \le \varepsilon$ is guaranteed by

$$N = O\left(\frac{\chi m}{\beta^2 c^2 \varepsilon^2} \log \frac{2m}{p}\right). \tag{37}$$

For root-mean-square coefficient error $m^{-1/2}\|\widehat{h} - h\|_2 \le \varepsilon$, the factor m in (37) disappears. This is not a coefficientwise ℓ_∞ guarantee. The measurement layer uses no entangling gates, $O(n)$ one-qubit operations per copy, and at most $O(nN)$ readout bits; streaming moment sums need $O(m)$ counters. An approximate optimizer with objective error ζ contributes at most $\sqrt{2\zeta/(\beta^2 c)}$ additional coefficient error.

9.2 Computation, precision and physical acquisition

The inverse theorem does not make F or its gradient inexpensive to evaluate. Computing them by dense diagonalization uses dimension 2^n, with $O(8^n + m4^n)$ arithmetic operations for a direct gradient evaluation and $O(4^n)$ dense working memory when observables are processed sequentially. These are illustrative arithmetic costs, not a bit-complexity advantage. Exponentials and gradients must be evaluated to controlled precision; an inexact-gradient method needs an error budget tied to $\beta^2 c\varepsilon$, not an exact oracle for free.

On a coefficient box, $\nabla^2 F \preceq \beta^2 m I$ follows from

$$\mathrm{Cov}_\rho^{\mathrm{BKM}}(A, A) \le \mathrm{Tr}(\rho A^2) \le \|A\|_{\mathrm{op}}^2 \le m\|u\|_2^2.$$

Hence standard projected-gradient optimization has a polynomial number of ideal gradient calls for fixed β, k, Δ, g, but those calls may themselves be exponentially expensive. No generic polynomial-time reconstruction algorithm is established by this argument.

Independent Gibbs-state copies are an input resource, not a freely preparable state. Equilibration time, access to equilibrium rather than a metastable state, readout noise and model mismatch require separate analysis. No total-resource superiority over the best arbitrary-temperature learning algorithms is claimed.

10 Targeted comparison with prior work

The 2021 free-energy convexity theorem of Anshu, Arunachalam, Kuwahara and Soleimanifar has an explicit $1/m$ loss on geometric lattices [1, Theorem 6 and its formal counterpart]. The bounded-degree appendix of *Learning SYK Hamiltonians*, available in October 2026, gives a $1/[m(1 + \log(2m))]$ lower bound for the same canonical bounded-incidence Pauli setting [7, Theorem D.1]. Theorem 1.1 removes these system-size losses in the stated inequalities. It is not a claim about dense SYK, whose incidence grows with system size.

High-temperature Hamiltonian learning was already made sample- and time-optimal by Haah, Kothari and Tang [2]. Rouzé and França give system-size-independent lower Hessian bounds for commuting Gibbs families [3, Appendix F.2]. The intended extension here is arbitrary finite temperature with genuinely noncommuting bounded-degree interactions, not rediscovery of those regimes.

Bakshi, Liu, Moitra and Tang already establish polynomial-time learning at every fixed temperature [4]. Chen, Anshu and Nguyen obtain near-optimal local learning on lattices and further bounded-degree results using KMS identities and operator Fourier transforms [5]. Bergamaschi's October 2026 stationarity framework proves convexity properties of a Fisher information defined through entropy production under a Gibbs sampler [6, Remark 1.3, Lemmas 3.3–3.4]. That functional is not the equilibrium BKM covariance matrix in Theorem 1.1.

The precise new assembly in this manuscript is: coefficient-ℓ_2 control of the joint tails, followed by the localized-adjoint image lemma, followed by a Cauchy–Schwarz transfer to the original equilibrium covariance. The local spectral bound, filtered-operator technology, and known positive special cases should each receive their separate provenance. The literature review was targeted rather than exhaustive; original priority remains unestablished.

11 Finite checks and remaining limits

The companion script `finite_checks.py` uses seed 261012. Six two- and three-qubit instances check the exact Gaussian modular identity, the pointwise KMS-adjoint identity, numerical integration of self-adjointness and positivity, and annihilation of the identity. Twenty-four additional instances use three- to five-qubit random noncommuting chains and antiferromagnetic Heisenberg cycles with small fields, at $\beta g = 0.2, 1, 4, 10$, and inspect the KMS and BKM covariance spectra. The star counterexample is also evaluated from its exact conditional distribution.

All declared finite tolerances passed. The largest pointwise adjoint relative error was approximately 1.03×10^{-15}; the smallest KMS eigenvalue in the covariance checks was approximately 2.53×10^{-5}. These checks neither determine the enormous worst-case constant nor validate the

uniform theorem, original priority, or practical performance. They are reproducible finite falsification checks only.

The global finite-dimensional anti-cancellation step is closed by the proof above under bounded candidate incidence. What remains outside this result is a sharp temperature constant, a corresponding theorem for growing-degree models under a different coefficient geometry, fixed-sector versions with their null directions removed, and an end-to-end computational advantage. Independent reconstruction of the proof and external priority assessment remain necessary before presenting this as an established new research contribution.

A Repository provenance

The public Astra research repository was inspected at revision

`3a9880bf4b1913a1b9e522f15414092e57c66eec`

and targeted searches in `openai/math` returned files at revision

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

No unreviewed theorem from either repository is assumed in this proof. In particular, a search route involving tensorization led to a different SLD-information question, not to a thermal covariance theorem. The earlier local thermal result is reconstructed in Section 2. No repository was modified during this investigation.

References

[1] A. Anshu, S. Arunachalam, T. Kuwahara, M. Soleimanifar. *Sample-efficient learning of quantum many-body systems*. Nature Physics (2021). arXiv:2004.07266. Inspected HTML v1, especially Theorem 6.

[2] J. Haah, R. Kothari, E. Tang. *Optimal learning of quantum Hamiltonians from high-temperature Gibbs states*. arXiv:2108.04842; FOCS 2022; related Nature Physics publication (2024).

[3] C. Rouzé, D. Stilck França. *Learning quantum many-body systems from a few copies*. Quantum 8, 1319 (2024). arXiv:2107.03333v4, Appendix F.2.

[4] A. Bakshi, A. Liu, A. Moitra, E. Tang. *Learning quantum Hamiltonians at any temperature in polynomial time*. arXiv:2310.02243, v2 (8 May 2026).

[5] C.-F. Chen, A. Anshu, Q. T. Nguyen. *Learning quantum Gibbs states locally and efficiently*. arXiv:2504.02706v1 (2025).

[6] T. Bergamaschi. *The stationarity test: a framework for learning quantum many-body systems from their thermal states*. arXiv:2610.01074 (October 2026), Remark 1.3 and Section 3.

[7] A. Anshu, S. Arunachalam, S. Chen, Y. Hwang. *Learning SYK Hamiltonians*. arXiv:2610.02178v1 (October 2026), Appendix D, Theorem D.1. The comparison here concerns that bounded-degree appendix, not the paper's dense-SYK theorem.

[8] C.-F. Chen, M. J. Kastoryano, A. Gilyén. *An efficient and exact noncommutative quantum Gibbs sampler*. arXiv:2311.09207v2 (2025).

[9] B. Nachtergaele, R. Sims. *Lieb–Robinson bounds and the exponential clustering theorem*. Communications in Mathematical Physics 265, 119–130 (2006). arXiv:math-ph/0506030v3.